Assignment Submission Guidelines

MBAV 443M • Statistical Modeling and AI-Driven Analytics

May – July 2026

Course Information

Item	Details
Course	MBAV 443M – Statistical Modeling and AI-Driven Analytics
Platform	Microsoft Teams
Repository	github.com/mbav443m
Course Email	mbav443m@gmail.com
Submission Guidelines	github.com/mbav443m/assignment-submission-guidelines

1. Overview

The course includes eight graded assignments distributed across the semester, each worth 10% of the final grade. The assignments are designed to develop analytical thinking, statistical modelling, data visualization, and professional communication skills through hands-on work with real-world datasets. Each student is assigned a unique dataset for each assignment to ensure that all submitted work reflects independent analysis.

Students are required to submit three components for every assignment: a concise analytical report, the supporting Jupyter Notebook, and the dataset used in the analysis. All files must be uploaded to the course GitHub repository before the stated deadline.

2. Submission Components

Component	Format	Where to submit
Analytical report	PDF	GitHub repository
Jupyter Notebook	`.ipynb`	GitHub repository
Dataset	`.csv` or source file	GitHub repository

3. Report Requirements

The analytical report must be between **two and three pages**. Reports shorter than two pages will be considered incomplete; content beyond page three will not be evaluated. This constraint is deliberate—concise, well-structured writing is a core professional skill and is assessed as part of every submission.

The report must follow the six-section structure provided in the assignment instructions, using the headings below in the order given. Write in complete prose sentences and paragraphs throughout. Do not include raw code, long output tables, or exploratory plots in the report body; these belong in the notebook.

1. Introduction
2. *[Assignment-specific analytical section—heading varies per assignment]*
3. Methodology
4. Visualization
5. Key Insight
6. Conclusions

The focus of the report should be on interpretation and communication of findings. Statistical outputs should be summarized and explained in plain language relevant to a business audience, not simply transcribed from code.

4. Notebook Requirements

The notebook must contain all code necessary to reproduce the analysis from start to finish. Organize it with Markdown headings that mirror the six report sections and include brief commentary between code blocks to explain what each step does. All cells must be executed and outputs must be visible before submission—do not submit a notebook with cleared outputs. The dataset file must be uploaded alongside the notebook so that the analysis can be re-run without modification.

5. File Naming Convention

All files must follow the naming convention below exactly, using your two-digit roll number and surname with no spaces.

File	**Example**
PDF report	`A1_07_Kumar.pdf`
Notebook	`A1_07_Kumar.ipynb`
Dataset	`A1_07_Kumar_data.csv`

Incorrectly named files may not be accepted. Replace `A1` with the assignment number, `07` with your roll number, and `Kumar` with your surname.

6. Submission Process

All files must be uploaded to the course GitHub repository in your designated folder before the announced deadline. Detailed upload instructions are available at the repository link listed in the course information table above. If a technical difficulty prevents timely repository submission, contact the course team at the course email address as early as possible with your files attached—do not wait until after the deadline. Extensions will not be granted after a deadline has passed; requests must be submitted in advance with a stated reason and a proposed alternate date.

7. Visualization Standards

Every report must include at least one visualization produced in Python using Matplotlib or Seaborn. The chart embedded in the report must be the final, polished version and must contain a descriptive title, labelled axes with units where applicable, and a legend if more than one series is displayed. Export figures at a minimum resolution of 150 DPI using `plt.savefig('figure.png', dpi=150)` and insert the saved image—do not paste screenshots. Exploratory or intermediate charts belong in the notebook, not in the report.

8. Academic Integrity

Students are expected to submit original work prepared independently. Each student is assigned a unique dataset specifically to ensure that analysis, code, and written interpretation cannot be copied. Discussion of concepts and approaches is encouraged; however, code, visualizations, and report text must be the student's own. Students may be asked to explain or reproduce their analysis during class, and inability to do so will be treated as evidence of academic dishonesty.

Generative AI tools may be used for learning, editing, and coding assistance. Students remain fully responsible for understanding, validating, and being able to explain all submitted work. If AI assistance was used in preparing a submission, briefly note this in the Conclusions section of the report.